**Project Design Phase**
**Solution Architecture**

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| Team ID | |
| Project Name | AI-Driven HealthCare Anomaly Detection |
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The solution architecture is a complex multi-layered pipeline designed to process high-frequency medical data with minimal latency. It bridges the gap between raw physiological sensors and clinical decision-making through integrated AI.

## CORE PROCESSES (SYSTEM BEHAVIOR)

Data Ingestion: Sub-processes handle the generation and streaming of JSON-formatted vital signs (Heart Rate, SpO2, Temp) via Apache Kafka.

AI Inference: An unsupervised Autoencoder model evaluates incoming data to determine a reconstruction error score.

Logic Enrichment: Detected anomalies trigger a request to the Gemini 1.5 Flash API to translate technical data into descriptive clinical insights.

## ARCHITECTURAL SPECIFICATIONS

Local Infrastructure: The Kafka broker, PostgreSQL database, and Flask web server are hosted on a local system configuration for high-speed data access.

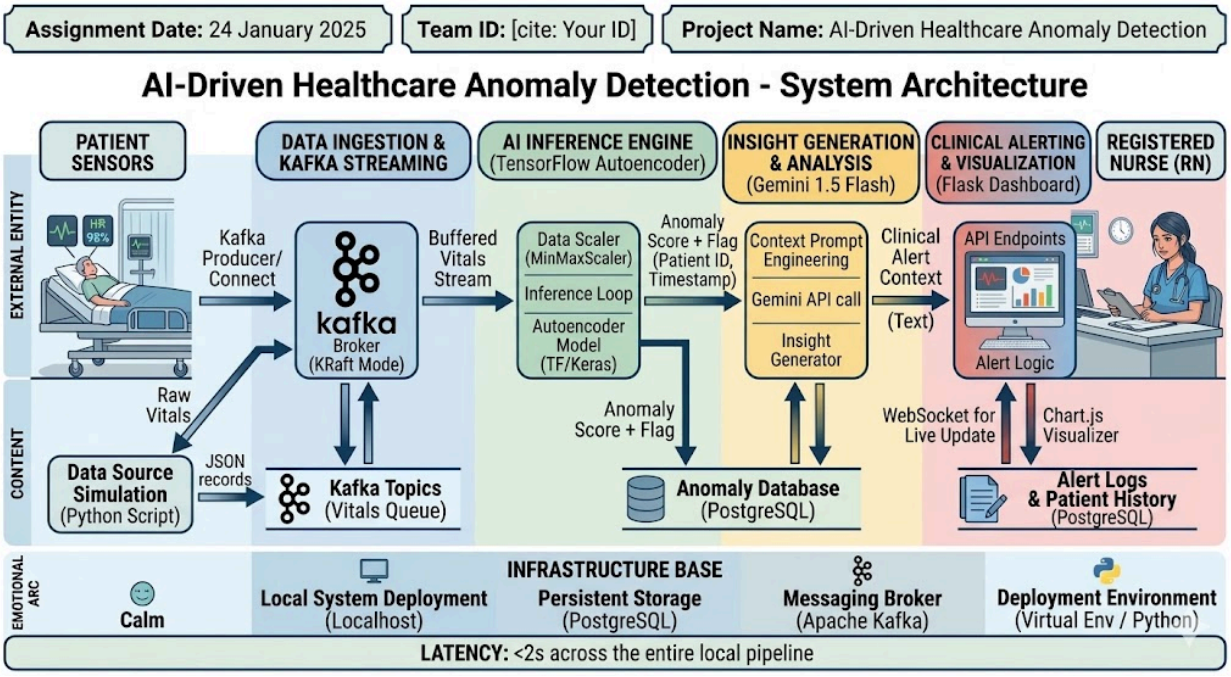
External Interfaces: The system utilizes a third-party API (Google Gemini) for advanced natural language processing.

Data Storage: A local PostgreSQL service ensures all processed patient data and AI-generated logs are persisted for long-term clinical review.

Frontend Interface: A real-time dashboard provides the user interface for stakeholders to visualize live patient health trends.

**DESIGN GOALS ACHIEVED**

Identification: Effectively finds the best technical solution to the problem of medical alarm fatigue.

Specification: Provides a clear technical blueprint from which the software was managed and delivered.

Stakeholder Value: Describes system behavior in a way that is transparent to clinical and technical stakeholders.